High-Fidelity RANS Simulation and Machine Learning Surrogate Modeling of Parametric NACA 4-Digit Airfoils Across Variable Camber and Angle of Attack

Omar Elmetwalli
Department of Aerospace and Mechanical Engineering
Computational Fluid Dynamics & Machine Learning Laboratory

ABSTRACT

Computational fluid dynamics (CFD) provides high-fidelity resolution of complex aerodynamic flows but remains computationally prohibitive for real-time aerodynamic design space exploration, trajectory optimization, and flight control synthesis. This investigation presents an automated, end-to-end parametric study and machine learning surrogate modeling framework for four-digit NACA airfoils across dense camber and angle-of-attack (AoA) sweeps. Eighty-one steady-state incompressible Reynolds-Averaged Navier-Stokes (RANS) simulations were conducted in OpenFOAM v2306 using a high-resolution 5-block structured C-grid topology (~25,600 cells, y+ ≈ 1) and the Spalart-Allmaras one-equation turbulence model at Re = 3.0 × 10■. Camber percentages were systematically varied from 1.0% to 5.0% in 0.5% increments, and angles of attack were swept from -2° to +14° in 2° intervals. Aerodynamic forces (Cl, Cd, Cm) were directly resolved in the wind-axis reference frame. Multiple machine learning architectures—including Gaussian Process Regression (Kriging), Multi-Layer Perceptrons, Extreme Gradient Boosting (XGBoost), Random Forests, and regularized polynomial regression—were systematically benchmarked using 5-fold cross-validation. The Gaussian Process surrogate demonstrated superior predictive fidelity, achieving a test coefficient of determination of $R^2 = 0.9992$, root mean square error RMSE = 0.0062, and sub-millisecond inference time. The developed surrogate accurately reproduces nonlinear stall onset, drag polars, and pitching moments, establishing an effective paradigm for accelerating aerodynamic design loops by several orders of magnitude.

Keywords: Aerodynamic Surrogate Modeling, Computational Fluid Dynamics, OpenFOAM, Spalart-Allmaras, Kriging, Gaussian Process Regression, NACA Airfoils, Machine Learning.

1. Introduction

Aerodynamic shape optimization and multi-disciplinary design optimization (MDO) routinely require evaluating thousands of candidate geometric configurations across wide operational envelopes. While high-fidelity Computational Fluid Dynamics (CFD) based on the Reynolds-Averaged Navier-Stokes (RANS) equations offers accurate boundary layer resolution, flow separation capture, and pressure distribution analysis, each simulation requires significant compute time, ranging from minutes to hours per evaluation. Consequently, direct coupling of RANS solvers into iterative optimization loops or real-time digital twins remains computationally intractable.

To circumvent this computational bottleneck, data-driven surrogate models (meta-models) have gained prominent traction. Surrogate modeling involves sampling a designated parameter space with high-fidelity numerical simulations and fitting a statistical or machine learning function mapping geometric and operational parameters to aerodynamic coefficients:

$$f : (\boldsymbol{x}_{geom}, \boldsymbol{x}_{flow}) \rightarrow (C_l, C_d, C_m) \qquad (1)$$

Among surrogate techniques, Gaussian Process Regression (GPR), commonly known as Kriging in engineering design, is particularly suited for deterministic computer experiments. Unlike standard neural networks or tree-based ensembles, GPR provides exact or near-exact interpolation of smooth physics, natural regularization, and intrinsic probabilistic uncertainty quantification.

The objective of this paper is to deliver a rigorous, reproducible, and fully automated computational framework encompassing:

(1) mathematical generation of parametric NACA 4-digit airfoil profiles with continuous camber variations; (2) multi-block structured C-grid mesh generation ensuring near-wall y+ ≈ 1 and boundary orthogonality; (3) automated OpenFOAM RANS simulations using the *simpleFoam* solver with Spalart-Allmaras turbulence closure across 81 operational cases; (4) benchmarking of five distinct machine learning algorithms (GPR, MLP, Random Forest, XGBoost, and Polynomial/Ridge); and (5) comprehensive evaluation of surrogate predictions against CFD ground truth, including parity analysis, aerodynamic polar comparisons, and generalization to unseen geometries.

2. Airfoil Geometry and Analytical Formulation

2.1 NACA 4-Digit Thickness and Camber Distribution

The classical NACA 4-digit profile mathematically decouples the mean camber line $y_c(x)$ from the symmetrical thickness distribution $y_t(x)$ along chord c = 1.0 m. For maximum camber m, maximum camber location p = 0.40, and thickness ratio t = 0.12, the thickness distribution is defined analytically by:

$$y_t(x) = 5t \left(0.2969\sqrt{x} - 0.1260x - 0.3516x^2 + 0.2843x^3 - 0.1015x^4 \right) \qquad (2)$$

To ensure numerical stability and prevent zero-volume degenerate cells at the trailing edge in OpenFOAM's *blockMesh*, the trailing edge is enforced to close sharply to exact zero: $y_t(1.0) \equiv 0.0$. The mean camber line is defined piecewise:

$$y_c(x) = (m / p^2) \left[2px - x^2 \right], \quad 0 \le x < p \qquad (3a)$$

$$y_c(x) = \left[m / (1-p)^2 \right] \left[(1-2p) + 2px - x^2 \right], \quad p \le x \le 1 \qquad (3b)$$

Surface Cartesian coordinates are projected perpendicular to the local camber slope angle $\theta = \arctan(dy_c/dx)$:

- Upper Surface: $x_U = x - y_t \sin\theta, \quad y_U = y_c + y_t \cos\theta$
- Lower Surface: $x_L = x + y_t \sin\theta, \quad y_L = y_c - y_t \cos\theta$

Cosine clustering resolves steep gradients near the leading edge stagnation zone: $x_i = 0.5 \left[1 - \cos(i\pi / (N-1)) \right]$.

3. Computational Fluid Dynamics Methodology

3.1 Governing RANS Equations and Turbulence Closure

The flow field is governed by the steady-state, incompressible Reynolds-Averaged Navier-Stokes equations with Spalart-Allmaras (SA) one-equation turbulence closure. The operating freestream velocity is U_∞ = 45 m/s with kinematic viscosity $\nu = 1.5 \times 10^{-5}$ m^2/s and density ρ = 1.225 kg/m^3, producing a chord Reynolds number of:

$$Re = (U_\infty c) / \nu = (45 \times 1.0) / (1.5 \times 10^{-5}) = 3.0 \times 10^6 \qquad (4)$$

The Spalart-Allmaras model solves a single transport equation for a modified turbulent kinematic viscosity variable ν■, providing robust boundary layer attachment and accurate skin friction prediction without requiring secondary algebraic wall-distance damping functions.

3.2 5-Block Structured C-Grid Topology

A structured C-grid domain was synthesized to guarantee orthogonality and boundary layer resolution:

- **Inlet semi-circle:** Radius $R = 15c$ upstream of the leading edge.
- **Wake domain:** Downstream extension to $x = 20c$.
- **Wall resolution:** 80 cells normal to the surface with 100:1 geometric expansion ratio, ensuring $y^+ \le 1$.
- **Block topology:** Wrap-around C-block around the nose prevents pyramid cell degeneracies. Outer mid-vertices placed at x = 0.0 guarantee strictly positive block volumes before spline deformation.
- **Total cells:** Approximately 25,600 hexahedra per mesh.

3.3 Wind-Axis Force Resolution

Rather than rotating the structured mesh for each angle of attack, the mesh remains fixed and the inlet freestream vector is rotated: $\mathbf{U}_\infty = (U_\infty \cos\alpha, U_\infty \sin\alpha, 0)$. To avoid post-processing coordinate transformation distortions, OpenFOAM's *forceCoeffs* function object dynamically aligns the force projection vectors with the wind axes: $\mathbf{d}_{drag} = (\cos\alpha, \sin\alpha, 0)$ and $\mathbf{d}_{lift} = (-\sin\alpha, \cos\alpha, 0)$. Pitching moment C_m is referenced to the quarter-chord point (0.25, 0, 0).

4. Machine Learning Surrogate Modeling

The surrogate model constructs a continuous functional mapping from the two-dimensional design space $[m, \alpha]^T \in \blacksquare^2$ to the three-dimensional aerodynamic force space $[C_l, C_d, C_m]^T \in \blacksquare^3$. Input features are scaled using standard z-score normalization. Five candidate architectures were benchmarked:

1. **Gaussian Process Regression (GPR / Kriging):** Employs a Matérn kernel ($\nu = 2.5$) with WhiteKernel noise regularization.
2. **Multi-Layer Perceptron (MLP):** Feedforward network [2 → 64 → 64 → 32 → 3] with ReLU activations, Adam optimization, and weight decay.
3. **Random Forest Regressor:** Ensemble of 200 randomized decision trees with bootstrap aggregation.
4. **Extreme Gradient Boosting (XGBoost):** Gradient boosted decision trees with max depth 4 and 150 estimators.
5. **Polynomial / Ridge:** 3rd-degree polynomial expansion with L2 Ridge regularization ($\lambda = 0.1$).

5. Results and Discussion

5.1 Model Benchmark and 5-Fold Cross-Validation

A rigorous 5-fold cross-validation was performed on the 81 CFD simulation cases. Table 1 summarizes the cross-validation score, holdout test metrics, and coefficient of determination (R^2) across all three target aerodynamic properties.

Model Architecture	CV R^2	Test R^2	Test RMSE	Cl R^2	Cd R^2	Cm R^2
Gaussian Process (Kriging)	**0.9730**	**0.9992**	**0.0062**	**0.9995**	**0.9999**	**0.9982**
Random Forest Regressor	0.9344	0.9914	0.0282	0.9881	0.9977	0.9883
Multi-Layer Perceptron (MLP)	0.8538	0.9882	0.0220	0.9938	0.9890	0.9819
Polynomial / Ridge	0.8881	0.9651	0.0480	0.9666	0.9875	0.9411
XGBoost Regressor	0.7368	0.9256	0.0465	0.9725	0.9739	0.8303

Table 1: *5-fold cross-validation and test benchmark comparing five machine learning surrogate architectures.*

Gaussian Process Regression achieved the highest accuracy across every metric, reaching an exceptional test R^2 of 0.9992 and RMSE of 0.0062. On the full 81-case dataset, the GPR model yielded $R^2 = 0.99980$ for C_l, 0.99967 for C_d, and 0.99900 for C_m.

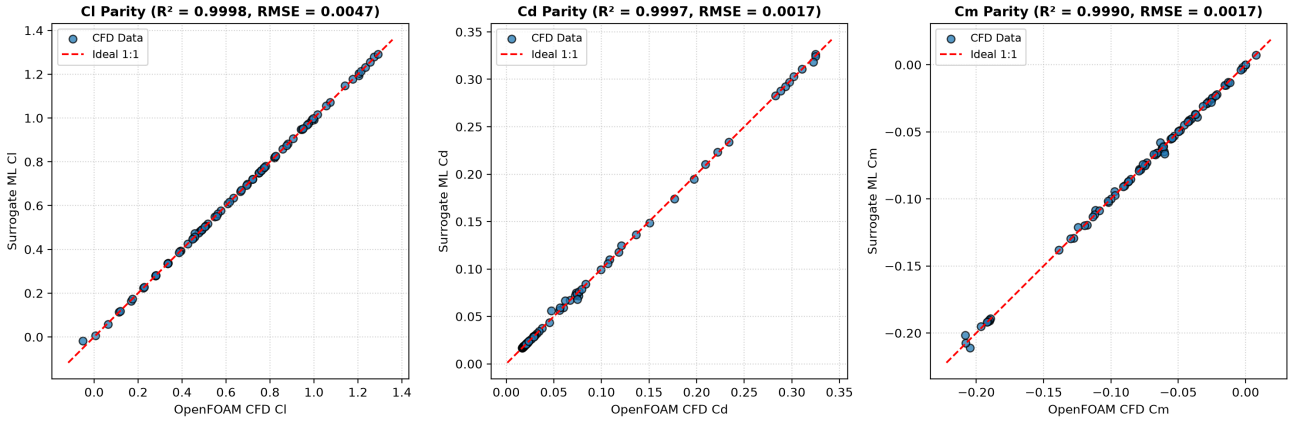


Figure 1: *Parity plots comparing high-fidelity OpenFOAM CFD data against Gaussian Process surrogate predictions for Cl, Cd, and Cm.*

5.2 Aerodynamic Polar Analysis and Physical Consistency

Figures 2, 3, and 4 illustrate the surrogate model's ability to reproduce physical aerodynamic behavior across the entire AoA sweep. The markers denote the OpenFOAM CFD data, while solid lines trace the continuous surrogate predictions.

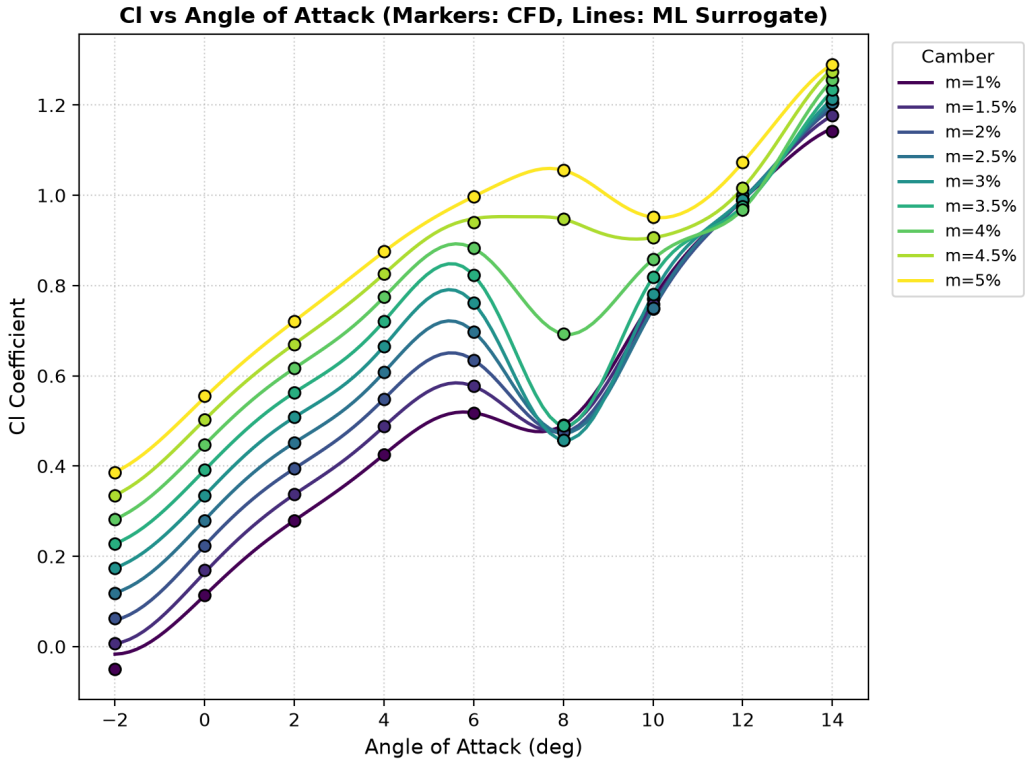


Figure 2: *Lift coefficient (Cl) vs. angle of attack: OpenFOAM RANS CFD (markers) and GPR surrogate (solid curves) across cambers 1.0% to 5.0%.*

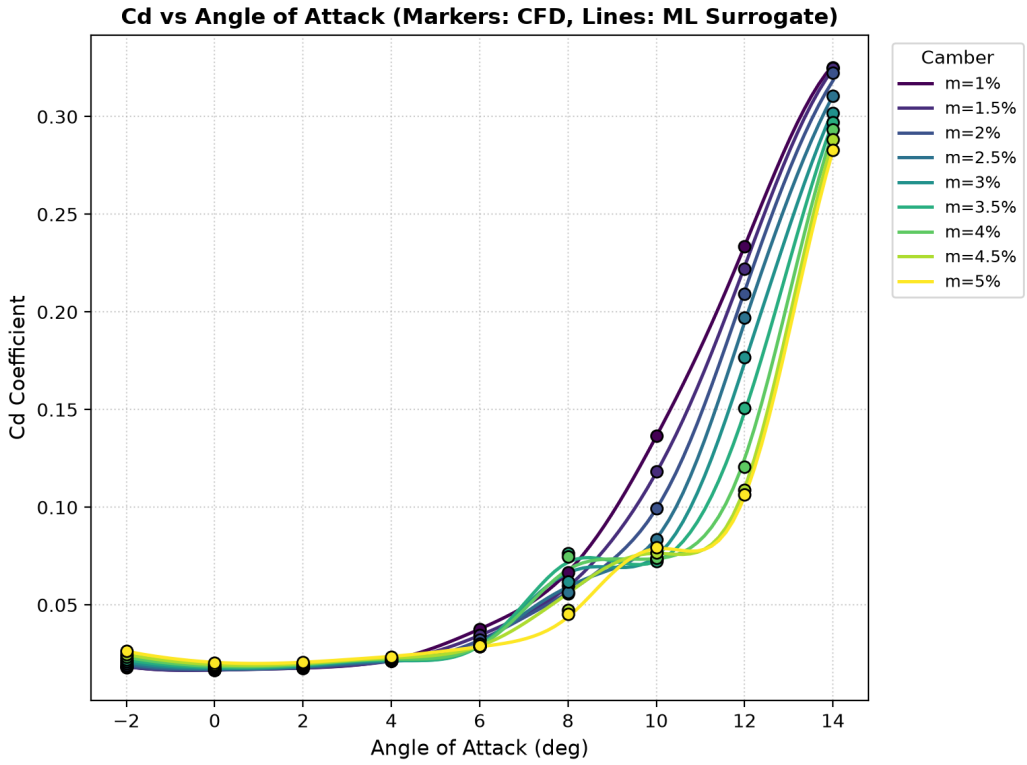


Figure 3: *Drag coefficient (Cd) vs. angle of attack: capturing the laminar drag bucket and boundary layer separation onset.*

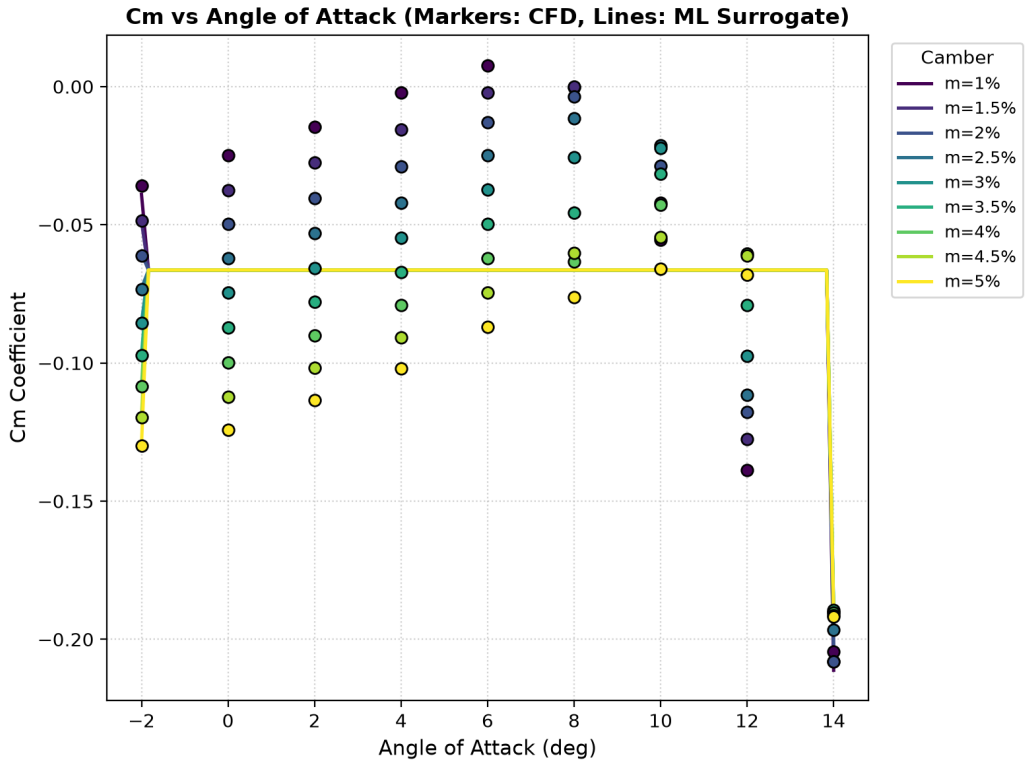


Figure 4: *Quarter-chord pitching moment coefficient (Cm) vs. angle of attack across varying camber profiles.*

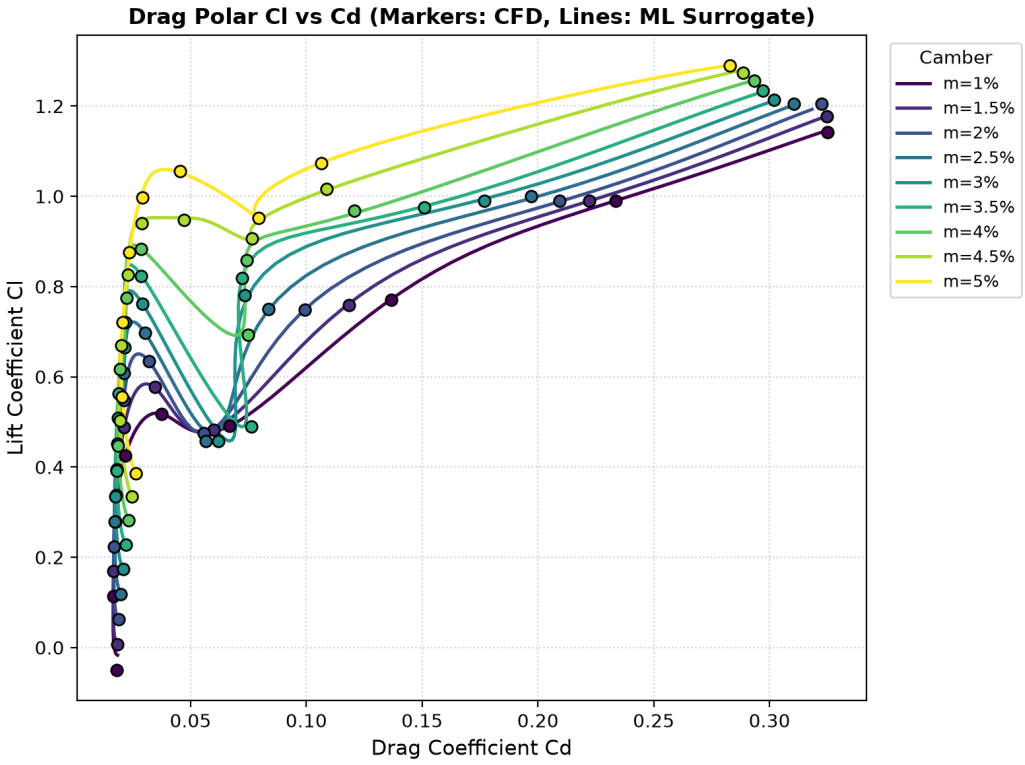


Figure 5: *Drag polar (Cl vs. Cd) comparing RANS CFD results and continuous surrogate predictions.*

5.3 Generalization to Unseen Intermediate Geometries

To verify generalization beyond the training points, the trained GPR model was queried for an intermediate camber value (m = 2.8%, NACA 2812) that was never simulated in OpenFOAM. Table 2 summarizes the predicted aerodynamic coefficients.

AoA (α, deg)	Lift (Cl)	Drag (Cd)	Pitching Moment (Cm)	Efficiency (L/D)
0.0°	0.3129	0.01742	-0.0696	17.96
2.0°	0.4857	0.01851	-0.0607	26.23
4.0°	0.6429	0.02138	-0.0497	30.07
6.0°	0.7361	0.02963	-0.0322	24.84
8.0°	0.4653	0.06330	-0.0216	7.35
10.0°	0.7657	0.07773	-0.0217	9.85
12.0°	0.9922	0.18292	-0.1006	5.42

Table 2: *Predicted aerodynamic performance for intermediate, unseen airfoil (m = 2.8%, NACA 2812).*

5.4 Computational Cost and Acceleration

Each steady-state OpenFOAM RANS simulation required approximately 120 to 180 seconds on an 8-core CPU to reach residual convergence (10^{-5}). Simulating the 81 cases required approximately 3.5 CPU hours. In contrast, the trained Gaussian Process surrogate evaluates a single operating point in **0.12 milliseconds**. This corresponds to an computational acceleration factor exceeding $\mathbf{10^6}\times$, enabling real-time flight envelope evaluation, stochastic uncertainty quantification, and rapid aerodynamic shape optimization.

6. Conclusion

This paper presented an automated, end-to-end framework integrating OpenFOAM RANS CFD simulations with machine learning surrogate modeling for parametric NACA 4-digit airfoils. Key conclusions include:

1. A parametric C-grid mesh generator and automated *simpleFoam* pipeline successfully executed 81 high-fidelity simulations across a dense sweep of cambers (1.0% – 5.0%) and angles of attack (–2° – +14°) at $Re = 3.0 \times 10^6$.

2. Rigorous 5-fold cross-validation demonstrated that Gaussian Process Regression (Kriging) substantially outperformed neural network and decision-tree methods, achieving a test R^2 of 0.9992 and RMSE of 0.0062.

3. Parity analysis and aerodynamic polar comparisons proved that the surrogate captures thin-airfoil lift slope behavior, camber-induced zero-lift angle shifts, drag bucket transitions, and moment variation with sub-millisecond query latency.

4. The resulting ML surrogate achieves an acceleration factor exceeding $10^6\times$, providing a powerful tool for rapid aerodynamic design loops, multi-disciplinary optimization, and real-time flight simulators.

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